TESTER PRESENT SPECIALIST AUTOMOTIVE SOLUTIONS

Tester Engineering Suite

Professional Automotive Diagnostic & ECU Reprogramming Platform

Version v1.0.1-2026

20+	40+	1000+	14
ECU FLASH PLATFORMS	SECURITY ALGORITHMS	OEM PIDS	UDS SERVICES

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KEY FEATURES OVERVIEW

20+ ECU FLASH PLATFORMS

100+ STANDARD PIDS

1000+ OEM PIDS

40+ SECURITY ALGORITHMS

14 UDS SERVICES

Multi-ECU Flash Platforms

Supports 20+ manufacturer-specific ECU reprogramming profiles covering Ford, VW/Audi, GM, Honda, Nissan, Mazda, Subaru, BMW, Mercedes, Saab, Volvo, Hyundai/Kia, Aston Martin, Jaguar Land Rover, Porsche and more.

20+ PLATFORMS

Full OBD-II Diagnostics

Complete Modes 01–0A support with 100+ standard PIDs and 1,000+ OEM-specific PIDs across all major manufacturers. Reads, clears, and logs fault codes with freeze-frame capture.

MODES 01–0A

UDS ISO 14229-1 Protocol

14 UDS services implemented including DiagnosticSessionControl, SecurityAccess, RoutineControl, RequestDownload, TransferData, ECUReset, ClearDTC, and more. Full tester-present keepalive.

14 UDS SERVICES

Security Access Algorithms

40+ built-in security algorithms covering manufacturer-specific seed-key derivation, AES-128, LFSR, RSA, EdDSA, and brute-force PIN attack routines for legacy and modern ECUs.

40+ ALGORITHMS

Real-Time Dashboard

10 configurable analogue-style gauges with live data streaming. Monitor RPM, coolant temp, MAF, throttle position, fuel trims, boost, lambda and any custom PID in real time.

10 LIVE GAUGES

CAN Bus Sniffer

Raw CAN frame capture and decode with live filtering, timestamping, and identifier lookup. Ideal for reverse-engineering custom frames and validating bus traffic during development.

REAL-TIME CAPTURE

Data Logging & Playback

Record multi-channel PID logs to CSV with configurable sample rates. Full playback engine allows review, annotation, and export of recorded drive sessions for diagnostics and tuning.

CSV EXPORT

J2534 PassThru Universal

Compatible with any SAE J2534-1 and J2534-2 compliant interface. Supports Mongoose, OpenPort 2.0, DrewTech, Tactrix, Cypress, Kvaser, and other third-party PassThru devices.

SAE J2534-1/2

Transmission Adaptation Control

Specialised ZF 6HP26 adaptation read/reset workflow. Reads A4 and A5 shift-point adaptation values and supports full calibration backup, reset file flash, and restore in one sequence.

ZF 6HP26

ROM & Calibration Management

Read and write full ROM images and 128 KB calibration files. Supports VBF, FRF, RWD, and BIN formats with automatic CRC-16 checksum validation and RSA signature verification.

VBF / FRF / RWD / BIN

SUPPORTED MANUFACTURERS & VEHICLES

Ford / Lincoln
PCM: Falcon BA–FG, Mustang GT/GT500, F-150 Coyote, Territory, Explorer
TCM: ZF 6HP26 & 5R55S
ELM / J2534

VW / Audi / SEAT / Skoda
ECU: Golf IV/V/VI, Polo, Passat, Tiguan, A3/A4/A6, TT, Ibiza, Octavia
Protocol: UDS + KWP2000
FRF / RWD

GM / Chevrolet / Holden
ECU: LS1/LS2/LS3, Commodore VT–VZ, Cruze, Colorado, Sierra, Tahoe
Protocol: J1850 VPW + UDS

Honda / Acura
ECU: Civic, Accord, CR-V, Jazz, Type R K-Series & H-Series, NSX
Protocol: KWP2000 + UDS

Nissan / Infiniti
ECU: 350Z/370Z, Skyline R32–R34, Silvia S13/S15, Patrol GU/GQ, Navara
Protocol: CONSULT-III+

Mazda
ECU: RX-8 (13B), MX-5 (NC/ND), Mazda 3/6/CX-5, BT-50
Protocol: UDS + KWP2000

Subaru
ECU: Impreza WRX/STI, Liberty GT, Forester XT, Outback H6
Protocol: SSM3 + UDS

BMW / MINI
ECU: E46/E90/E60/E70, M3/M5/X5, MSS54/MSS65, MSx DME
Protocol: EDIABAS + UDS
MSX SERIES

Mercedes-Benz / AMG
ECU: C/E/S/G-Class, SL, CLA/GLC, AMG 63, Sprinter
Protocol: DAS/Xentry UDS

Saab
ECU: 9-3/9-5, Trionic T5/T7/T8 ECUs
Protocol: Trionic + KWP2000
TRIONIC

Volvo
ECU: S40/S60/V70/XC60/XC90, D2/D4/D5/T6/T8 engines
Protocol: VIDA + UDS
D-SERIES

Hyundai / Kia / Genesis
ECU: i30N, i20N, Elantra GT, Sonata, Tucson, Sportage, Stinger GT
Protocol: UDS + KWP2000

Aston Martin
ECU: Vantage, DB9, DBS, Rapide — V8 Vantage & V12 engines
Protocol: UDS + J2534
V12 VANTAGE

Jaguar Land Rover
ECU: F-Type, XE/XF, Range Rover Sport, Discovery, Defender 2.0/3.0
Protocol: SDD + UDS

Porsche
ECU: 911 (991/992), Cayman, Macan, Cayenne, Panamera
Protocol: PIWIS + UDS
PIWIS

Toyota / Lexus
ECU: Hilux, LandCruiser, Supra A90, Camry, IS/GS/LS
Protocol: Techstream + UDS

Mitsubishi
ECU: Lancer EVO VII–X, Eclipse, Outlander, Triton
Protocol: MUT-III + UDS

Chrysler / Dodge / Jeep
ECU: Challenger, Charger, Wrangler JK/JL, Grand Cherokee, RAM
Protocol: J1850 PWM + UDS

SPECIALTY & EXOTIC SUPPORT HIGHLIGHTS

Aston Martin V12	Saab Trionic	Volvo D-Series	BMW MSx DME
Vantage / DB9	T5 / T7 / T8	D2 / D4 / D5	MSS54 / MSS65
UDS + J2534	9-3 & 9-5	Diesel ECU	M3 / M5 / M6

TECHNICAL CAPABILITIES

FLASH PROTOCOLS

- CAN ISO 15765-2
- K-Line ISO 14230
- J1850 VPW
- J1850 PWM
- Dual-Channel CAN
- 9-bit McMESS
- KWP2000 Fast
- KWP2000 Slow
- ISO 9141-2
- UDS over CAN
- TP 2.0 (VAG)

SECURITY & CRYPTOGRAPHY

- AES-128
- LFSR Seed-Key
- RSA Signature
- EdDSA
- Brute-Force PIN
- XOR Cipher
- Polynomial LFSR
- Manufacturer Seed-Key x40+

FILE FORMAT SUPPORT

- VBF (Volvo)
- FRF (VAG)
- RWD (VAG)
- BIN / HEX
- SREC (Motorola)
- Intel HEX

UDS SERVICES (ISO 14229-1)

- 0x10 DiagnosticSessionControl
- 0x11 ECUReset
- 0x14 ClearDiagnosticInfo
- 0x19 ReadDTCInfo
- 0x22 ReadDataByID
- 0x23 ReadMemByAddr
- 0x27 SecurityAccess
- 0x28 CommControl
- 0x2E WriteDataByID
- 0x31 RoutineControl
- 0x34 RequestDownload
- 0x36 TransferData
- 0x37 RequestTransferExit
- 0x3E TesterPresent

CHECKSUM & VALIDATION

- CRC-16 CCITT
- CRC-32
- RSA Signature Verify
- ROM Size Validation
- Header Magic Check
- Pre-flash Verification

OBD-II DIAGNOSTIC MODES

- Mode 01 — Live PIDs
- Mode 02 — Freeze Frame
- Mode 03 — Read DTCs
- Mode 04 — Clear DTCs
- Mode 05 — O2 Monitor
- Mode 06 — Monitor Results
- Mode 07 — Pending DTCs
- Mode 08 — Actuator Tests
- Mode 09 — Vehicle Info
- Mode 0A — Perm DTCs

SAFETY & FLASH INTEGRITY FEATURES

- ✓ Pre-flash communication sequence validation
- ✓ Automatic tester-present keepalive during flash
- ✓ Post-flash ROM verification read-back
- ✓ TransferData block CRC validation per chunk
- ✓ Negative response code (NRC) full decode
- ✓ Battery voltage monitoring (min. 12.5 V)
- ✓ Mid-flash abort with safe ECU state exit
- ✓ Calibration backup before any reset operation
- ✓ Session timeout detection and auto-recovery
- ✓ Flash progress with byte-level reporting

SYSTEM REQUIREMENTS

COMPONENT	MINIMUM	RECOMMENDED
Operating System	Windows 10 (64-bit)	Windows 10 / 11 Pro (64-bit)
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5+ / Ryzen 5+
RAM	4 GB	8 GB or more
Storage	500 MB free (installation)	2 GB+ (calibration library)
Display	1280 x 720	1920 x 1080 or higher
.NET Runtime	.NET Framework 4.8	.NET Framework 4.8 / .NET 8
Interface Port	USB 2.0	USB 3.0 (low-latency)
J2534 Driver	SAE J2534-1 compliant	J2534-1 + J2534-2 (dual channel)
Vehicle Power	12.0 V (engine off)	12.5 V+ (battery charger recommended)

SUPPORTED J2534 DEVICES

Mongoose Pro	OpenPort 2.0	DrewTech CarDAQ	Tactrix OpenPort
Cypress AT2004	Kvaser Leaf	PEAK PCAN-USB	Vector CANcase
Bosch VCI	Autel MaxiFlash	Drew Tech Mongoose	+ Any J2534-1/2

Get in Touch

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